

COURSE 3 - DATA ENGINEERING 2026

C3-081 - Cloud & Microsoft Fabric Data Engineering Practice Pack

Genuine-attempt tasks. Live Fabric evidence is required only where the task makes a live platform claim.

Artifact	C3-081
Status	2026 - LEARNER EDITION
Phase	C3-07 Cloud & Microsoft Fabric Data Engineering 15 lessons 50-70 focused hours
Prerequisite	C3-06 Gate PASS; C2B cloud foundations are recalled, while Fabric implementation is taught from first principles.
Current lane	Microsoft Fabric docs refreshed 2026-09-11 trial 60 days F4/F64 up to 1 TB OneLake
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided Fabric action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> security/production reasoning -> evidence. A portal screenshot is not mastery unless the underlying data, identity, lifecycle and recovery claim is independently verified.

Attempt protocol

For each Pxx: predict the platform/data/identity outcome, complete the matching starter artifact, save evidence/uncertainty, then open only the matching protected review. Local reference scripts are controls, not substitutes for tenant execution.

Starter ZIP

Extract C3_081_PRACTICE_STARTER_PROJECT_RC1.zip. Keep secrets and environment-specific IDs out of committed evidence.

P01 - Cloud data-platform primitives: storage, compute, network and IAM

Draw a source -> Fabric movement -> Lakehouse -> Warehouse/BI architecture and label storage, compute, network/control and identity.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
- Independent row/key/total/security/monitoring validation appropriate to the task
- One break/fix or blocked-access result
- Confidence 1-4 + main uncertainty
- 60-90 second own-words explanation

After saving this genuine attempt, C3-082 R01 may be opened.

P02 - Identity, RBAC, secrets and least privilege

Complete a role/identity matrix for engineer, analyst, pipeline identity and external vendor. Include one blocked consumer test and secret boundary.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
- Independent row/key/total/security/monitoring validation appropriate to the task
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- Confidence 1-4 + main uncertainty
- 60-90 second own-words explanation

After saving this genuine attempt, C3-082 R02 may be opened.

P03 - Fabric workspaces, OneLake, shortcuts and mirroring architecture

Classify four sources as shortcut, mirroring, Copy Job/pipeline movement or native placement, with ownership/freshness/recovery reasons.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
- Independent row/key/total/security/monitoring validation appropriate to the task
- One break/fix or blocked-access result
- Confidence 1-4 + main uncertainty
- 60-90 second own-words explanation

After saving this genuine attempt, C3-082 R03 may be opened.

P04 - Fabric lakehouse and Delta tables

Define Bronze Files/Delta and Silver Delta contracts for orders. Include grain, key, schema, idempotent correction and reconciliation.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
- Independent row/key/total/security/monitoring validation appropriate to the task
- One break/fix or blocked-access result
- Confidence 1-4 + main uncertainty
- 60-90 second own-words explanation

After saving this genuine attempt, C3-082 R04 may be opened.

P05 - Data Factory pipelines, Copy activity and Dataflow Gen2 decision

Choose among Copy Job, Copy activity/pipeline, Dataflow Gen2 and notebook/Spark for five movement/transformation cases; complete the pipeline TODO.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

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- Independent row/key/total/security/monitoring validation appropriate to the task
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- Confidence 1-4 + main uncertainty
- 60-90 second own-words explanation

After saving this genuine attempt, C3-082 R05 may be opened.

P06 - Connections, parameters and gateway boundaries

Complete connection/parameter/gateway decisions for SaaS API, public cloud storage, on-prem SQL and private endpoint scenarios.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
- Independent row/key/total/security/monitoring validation appropriate to the task
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- Confidence 1-4 + main uncertainty
- 60-90 second own-words explanation

After saving this genuine attempt, C3-082 R06 may be opened.

P07 - Fabric notebooks with PySpark

Complete the PySpark notebook TODO with explicit schema, reusable transform, Delta target and independent count/amount validation.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
- Independent row/key/total/security/monitoring validation appropriate to the task
- One break/fix or blocked-access result
- Confidence 1-4 + main uncertainty
- 60-90 second own-words explanation

After saving this genuine attempt, C3-082 R07 may be opened.

P08 - Spark job definitions, environments and scheduling

Complete Spark Job Definition TODO and Dev->Test promotion contract, including environment/default-lakehouse remapping and smoke evidence.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
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- Independent row/key/total/security/monitoring validation appropriate to the task
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After saving this genuine attempt, C3-082 R08 may be opened.

P09 - Fabric Warehouse and T-SQL serving layer

Complete Warehouse schema + serving-view TODOs with explicit fact grain, dimensions, T-SQL serving contract and reconciliation.

Before you build

Write: source + target grain/resource boundary + acting identity + expected success evidence + independent validation + one plausible failure.

Evidence to save

- Your changed file/definition/query/diagram
- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
- Independent row/key/total/security/monitoring validation appropriate to the task
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After saving this genuine attempt, C3-082 R09 may be opened.

P10 - Eventstreams, Eventhouse and KQL foundations

Write the KQL TODO for site-level five-minute telemetry and draw Eventstream -> Eventhouse flow; state why Warehouse is different.

Before you build

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Evidence to save

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- Observed Fabric run/query/security evidence when available, or explicit PENDING note if not
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After saving this genuine attempt, C3-082 R10 may be opened.

P11 - Incremental and CDC-style implementation patterns

Run the local reference-controls script on initial+incremental orders, prove one update + one insert, safe rerun and durable watermark design.

Before you build

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Evidence to save

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After saving this genuine attempt, C3-082 R11 may be opened.

P12 - Git integration, database projects and deployment pipelines

Write branch/PR/deployment plan. Explicitly state which Lakehouse data is not versioned by Git and which references must be remapped.

Before you build

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Evidence to save

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After saving this genuine attempt, C3-082 R12 may be opened.

P13 - Monitoring and performance optimization

Complete monitoring matrix for pipeline, Spark job, Warehouse/Eventhouse and capacity. Every signal needs detect -> diagnose -> recover -> validate.

Before you build

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Evidence to save

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After saving this genuine attempt, C3-082 R13 may be opened.

P14 - Fabric security & governance: access scopes, masking, labels, audit and OneLake security

Complete a security matrix using control-plane role plus OneLake data-plane role/RLS/CLS. Test with the actual Viewer/consumer identity.

Before you build

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Evidence to save

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After saving this genuine attempt, C3-082 R14 may be opened.

P15 - Capacity/cost awareness and DP-700 objective mapping

Build a capacity-aware weekly schedule and DP-700 evidence map. Explain how you would react to CU pressure without treating the exam map as mastery.

Before you build

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Evidence to save

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After saving this genuine attempt, C3-082 R15 may be opened.